

Sustainable management of planted landscapes: lessons from Japan

Yuichi Yamaura · Hiroyasu Oka · Hisatomo Taki · Kenichi Ozaki · Hiroshi Tanaka

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Abstract In Japan, 42 % of forests are planted forests, and most of them were established after World War II (1950–1980) to meet increased wood demands. Although Japanese planted forests are now reaching their planned harvest age, they have not been managed, and their restoration is now being discussed. Japanese foresters have not cut their own forests, and the country's high wood demands have been met by imports during recent decades. The decline of young forests due to the stagnation of forestry activity is suggested to be partly responsible for the nation-wide decline in early-successional species, which is referred to as the “second crisis of biodiversity.” As a timber-importing nation, it is suggested that Japan has underused the nation's own forests and has overused forests elsewhere. A revival of Japanese plantation forestry may contribute to the restoration of early-successional species because young planted forests are likely to provide suitable habitats. Furthermore, only 30 % of the current planted forests in Japan will be needed to meet the expected future domestic demand for lumber and plywood without imports. The remaining 70 % of the current planted forests may be restored to natural forests with or

Y. Yamaura (✉)

Division of Environmental Resources, Graduate School of Agriculture, Hokkaido University,
Kita 9 Nishi 9, Kitaku, Sapporo, Hokkaido 060-8589, Japan
e-mail: yamaura@for.agr.hokudai.ac.jp

H. Oka

Department of Forest Policy and Economics, Forestry and Forest Products Research Institute,
1 Matsunosato, Tsukuba, Ibaraki 305-8687, Japan

H. Taki

Department of Forest Entomology, Forestry and Forest Products Research Institute, 1 Matsunosato,
Tsukuba, Ibaraki 305-8687, Japan

K. Ozaki

Hokkaido Research Center, Forestry and Forest Products Research Institute, Hitsujigaoka-7, Toyohira,
Sapporo, Hokkaido 062-8516, Japan

H. Tanaka

Department of Forest Vegetation, Forestry and Forest Products Research Institute, 1 Matsunosato,
Tsukuba, Ibaraki 305-8687, Japan

without harvesting. The history of Japanese planted forests suggests that some natural trees/forests should be retained, even in the landscapes that specialize in wood production, because part of the planted forests may be economically marginalized in the future, and their restoration to natural forests would then be needed.

Keywords Economic marginalization · Forest planning · Forest restoration · Forest-use history · Overplanted forests · Planted forest expansion · Retention of natural forests

Introduction

World planted forests expanded 23 % between 2000 and 2010 (FAO 2010). Although these forests represent only 6.6 % of the global forested area (264 million ha; FAO 2010), their contribution to global wood production approaches 50 % (FAO 2007). The expansion of planted forests will probably continue or accelerate (e.g., Puyravaud et al. 2010), partly due to carbon sequestration policies (Canadell and Raupach 2008). Planted forests typically consist of one or a few tree species that are regularly spaced and even-aged (Carnus et al. 2006). Devastating impacts on animal communities can occur when natural forests are converted to planted forests (e.g., Nájera and Simonetti 2010). Planted forests can dominate the landscape as a matrix, and natural forests are retained as remnant fragments (Lindenmayer et al. 2002), which are usually restricted to less productive or less commercially valuable areas (e.g., upland mountainous areas; Yamaura et al. 2011). Loss and fragmentation of natural forests leads to lower species diversity within the remaining natural forest (e.g., Lindenmayer et al. 2002; Yamaura et al. 2009b). These studies invoke the notion that planted forests themselves are homogeneous non-habitats, i.e., “green deserts” (sensu Koh and Gardner 2010). However, it is not simply the existence of planted forests but the kind of planted forests that is important. For example, Japanese planted forests can harbor as large a number of species as natural forests, and the extent depends on local stand structure (Ito et al. 2003; Nagaike et al. 2006; Yamaura et al. 2008a), surrounding landscape structure, (Utsugi et al. 2006; Yamaura et al. 2006), and previous land use (Ito et al. 2004; Sugita et al. 2008). This suggests that the potential exists to maintain biodiversity in landscapes dominated by planted forests. While ongoing global expansion of planted forests increases the need for biodiversity conservation in planted landscapes (Brockerhoff et al. 2008; Paquette and Messier 2010), several countries, such as Japan and Great Britain, have already experienced a substantial expansion of planted forests (Totman 1989; Kirby 1992). Here, we briefly look back at these histories (mainly in Japan), discuss their implications for biodiversity conservation, and suggest that the experiences of countries that developed plantations, including Japan, can and should provide important information for biodiversity conservation in landscapes dominated by planted forests worldwide.

History of forest use and planted forests in Japan

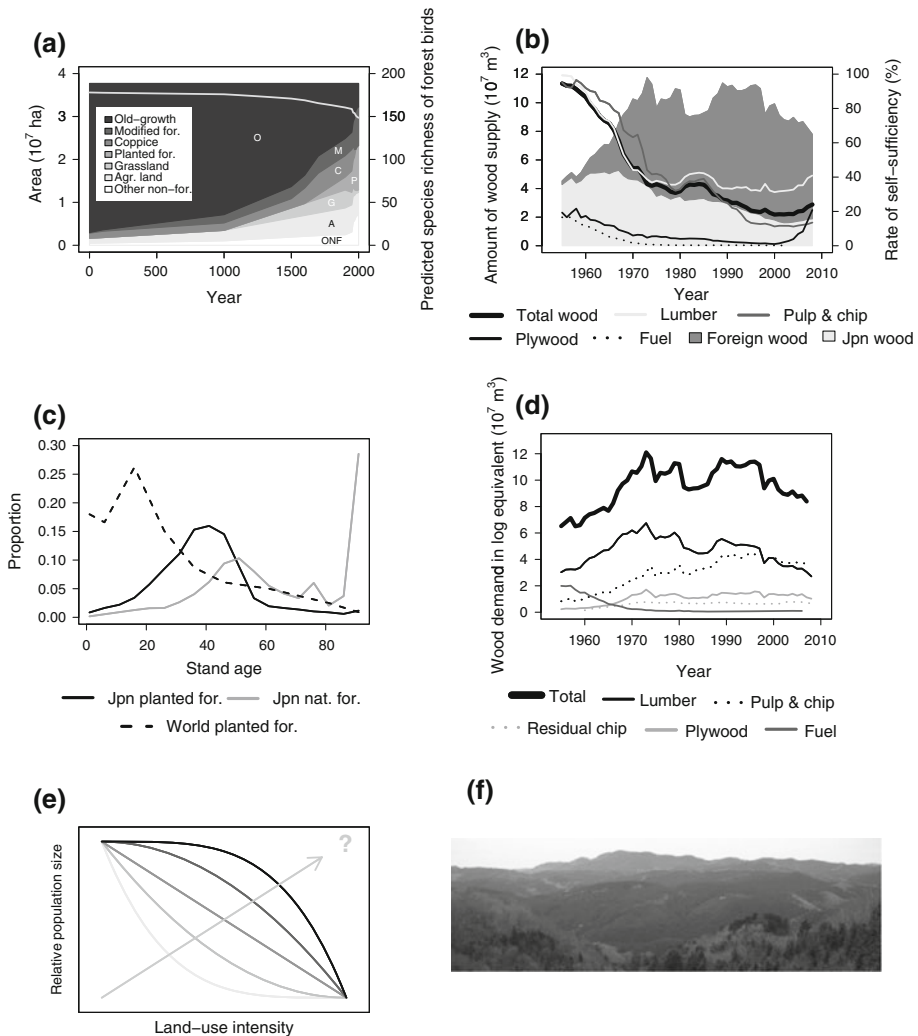
The earliest significant human pressure on Japanese forests occurred in the eleventh century (Yorimitsu 1984). The population of Japan increased to 10 million, and forests were harvested and converted to grasslands for agricultural use (Fig. 1a). From the beginning of the Peace Era (Edo period; early seventeenth century), the pressure on forests greatly increased, and planted forests were created by local authorities (e.g., feudal

Fig. 1 Status and social demands on Japanese forests **a** Land-use history of Japan. Data from Yorimitsu (1984), the Japanese Ministry of Internal Affairs and Communications (<http://www.stat.go.jp/english/index.htm>), and Himiyama (1995). A modified natural forest is a natural forest with clear visible human activities (including saw-timber natural forest and abandoned coppice: FAO 2007). The gray line indicates the predicted species richness of breeding Japanese forest birds via land-use change using the method of Koh and Ghazoul (2010) (Appendix Table 1). The initial forest species richness was estimated at 178 species versus the current 170 species (8 extinctions), according to Tojo (2007). In 2000, species richness was estimated at 148, with about 20 species estimated to be extinct in the future (i.e., living dead). **b** Wood supply of Japan. Wood supply amount and self-sufficiency rate are shown by areas and lines, respectively, except for fuel supply, which is indicated by a dotted line and is not included in the wood supply (i.e., foreign and Japanese wood) or self-sufficiency rate. Source: Forestry Agency (2010), except for the 1955–2002 fuel supply, which is from wood supply and demand data from the Ministry of Agriculture, Forestry, and Fisheries. **c** Present age-class distribution. Note that resolutions of age-class distributions of Japanese and world forests differ. Sources: Japanese forests (Status of Forest Resources, Forestry Agency, Japan: <http://www.rinya.maff.go.jp/toukei/genkyou/index.htm>); world planted forests (FAO 2006). The upper age class includes >91-year-old forests. **d** Wood demand in Japan. Source: wood supply and demand (Ministry of Agriculture, Forestry, and Fisheries, Japan: <http://www.maff.go.jp/e/index.html>). **e** Possible relationships between population size and land-use intensity. Retention of keystone structure and land-use configuration may change these relationships. **f** Remnant belt of natural deciduous forests in an evergreen planted landscape (photographed after leaf fall for contrast)

domains, villages) and individuals. The population reached 30 million at that time, and forests were cut to meet the increased demand of big cities for wood materials and fuels. Large areas were grassland in Japan during this period (more than 10 % of the total land). Coppice management was intensified to produce fuel and agricultural fertilizer. Forest use was further intensified in the late nineteenth century (Meiji period). Commercialization, infrastructure, and transportation systems developed rapidly at that time, and grasslands began to be converted into tree plantations.

The most drastic changes in forest utilization occurred after the 1950s in Japan. Grasslands and coppice forests were abandoned because of the introduction of new fuels (coal and oil) and the development of chemical fertilizers. Wood demands for construction increased, and the Japanese government began to promote a program to increase timber production by replacing grasslands, coppice forests, and modified natural forests with highly productive coniferous planted forests. Planted forests now account for more than 40 % of the forests, and grasslands and natural forests have greatly decreased (Fig. 1a). Grasslands now occupy only 1–2 % of the land. Substantial decreases in natural forests in the late twentieth century may have resulted in partial extinctions of Japanese forest birds (Fig. 1a). The three main planted species, Japanese cedar *Cryptomeria japonica*, Hinoki cypress *Chamaecyparis obtusa*, and larch *Larix kaempferi*, are all native to Japan but were widely planted outside their natural distributions. Large amounts of natural broad-leaved forests, which were mainly composed of *Fagaceae* and *Lauraceae* tree species, were replaced by these planted forests.

Notably, recent paleoecological studies suggest that grasslands have been a major ecosystem in Japan for tens of thousands of years (Yumoto 2011; Ogura 2012; Suka et al. 2012). Before 13,000 years ago (i.e., during the last glacial stage), cool and dry climates had prevented forests from developing, and grassland prevailed in many parts of Japan. After the last glacial stage, grasslands were maintained by human activities such as periodic burning and heavy forest cutting. The Japanese people of that time needed large areas of grassland for use in hunting, as pasture for horses, and in the production of natural fertilizers. Deep soil samples, old pictures, maps, and paleogeographies indicate the existence of vast grasslands in Japan at that time. Black soils, which are considered to be formed from volcanic ash under grassland, cover 17 % of Japan. The distribution of black soils



well coincides with the distribution of two major forest types in eastern Japan (cool-temperate mixed forests and warm-temperate deciduous forests), suggesting that these forests may have established in the past grasslands (Nakashizuka and Iida 1995). Grassland should be re-evaluated as a historically more important land use than was previously recognized in Japan. It is also suggested that man-made Japanese grasslands functioned as a remnant refuge for grassland species after the last glacial stage (Moriyama 1988; Suka et al. 2012).

By replacing such grasslands and natural forests with planted forests, Japan now has the fifth-largest area of planted forests in the world (10.3 million ha), and its proportion of planted forest to total forested area is the second largest in the world (42 %; the first is Great Britain; FAO 2006). Quite high proportions of planted forests are found in southwestern Japan; e.g., 65–66 % of the total forested area is planted forest in Kochi and Saga prefectures, respectively (Forestry Agency 2007b). After the 1960s, Japanese forestry

gradually declined with the increase in imported foreign wood and timber (Fig. 1b), and most of the expanded planted forests have not been managed since that time (Forestry Agency 2010). Although many of the planted forests are now reaching harvest age (Fig. 1c), the owners are reluctant to harvest their forests, and wood supply from the planted forests has only slightly increased. Annual log production of conifer is recently <0.5 % of the growing stock of planted forests. And not more than 2 % of the forests are younger than 10 years old (Forestry Agency 2007b). Modification of forestry activities in these planted forests, including rotation age extensions and rezoning of areas suitable for economic forest management, has been discussed by the forestry community in Japan (e.g., Forestry Agency 2010). Ironically, reconversion of planted forests into natural forests is currently being discussed, despite the immense effort required to establish planted forests in the past. For example, the Forestry Agency (2009) included the reconversion of planted forests into the future strategy of biodiversity conservation in forests.

Planted forests have been reconverted to natural forests in some European countries (Brockerhoff et al. 2008). For example, many remaining natural forests, including coppices and ancient forests of rural Great Britain, were replaced by planted coniferous forests after World War II (since the 1950s), but a movement to restore the natural forests started in the 1980s due to the concern for the loss of these semi-natural areas (Kirby 1992). Another analogy is the recent abandonment of agricultural fields in the tropics, followed by the regeneration of secondary forests in hilly, upland, and mountainous areas due to the marginal agricultural value of those areas (Wright and Muller-Landau 2006; Hecht and Saatchi 2007; Asner et al. 2009). Agricultural fields have been on the decrease in Japan since the 1950s (Fig. 1a), as in certain other regions of the world (Ellis et al. 2010). Although land-use transitions depend on the local environment, historical background, and the international economy (Perfecto and Vandermeer 2010), we suggest that some of the planted forests now being developed worldwide may face the same kind of problems occurring in Japan and Great Britain.

Current issues: how should overplanted forests be managed?

We believe that many countries and regions, where planted forests are now expanding can learn from Japan's historical experiences and current challenges with its planted forests. The biggest and most urgent issue is the long-term management of overplanted forests. Japan's self-sufficiency rate for wood supply has been around 20 % in recent decades (Fig. 1b), and a large portion of Japan's vast planted forests have not been managed. It can be assumed that 70 % of planted forests are targeted for timber harvest. If it is further assumed that a maximum harvest age is set to 160 years, and the skewed age distributions of planted forests are considered, at least 60,000 ha of planted forests need to be cut each year in average (Appendix 2). Such a cut area, which is 1.7 times the current cut area, would be sufficient to meet present domestic demands for lumber and plywood (Appendix 2), although it should be noted that the demand for lumber and plywood has been decreasing in the last few decades (Fig. 1d), and the Japanese population has also started to decrease (it may be reduced to less than 40 % by 2110 compared with 2010; Department of Population Dynamics Research 2012). Even assuming that future wood consumption per capita does not change, the area of productive planted forest required to meet future domestic demand for lumber and plywood is only about 30 % of the existing area (Appendix 3).

These rough estimates suggest that restoration of natural forests after harvesting planted forests may often be an adequate decision which do not harm the interests of future

generations. A possible future scenario for Japanese planted forests is partial reconversion to natural forests after harvesting. However, forest owners often feel guilty about harvesting without subsequent replanting. Furthermore, because the present cost of regeneration by planting is often estimated to be higher than the net revenue from the final harvest of the planted forest, most forest owners are reluctant to harvest planted forests. As a result, planted forests may become long-term negative assets that do not generate money, but instead cost money. Large amounts of planted forest may be degraded before harvest in terms of wood production, where native tree species can establish in old planted forests (Ohsawa and Shimokawa 2011), simple abandonment may be another method to restore planted forests.

Here, we need to draw a distinction between areas that should be restored to natural forests and those that should be replanted after harvest. With the expectation that natural forests could be restored by natural regeneration after clear-cutting of planted forests, some forest owners have harvested their mature planted forests without replanting. Recent studies have indicated that little erosion, including few landslides, had occurred in abandoned clear-cut stands by several years after the harvest (Kajisa et al. 2011). In terms of tree regeneration success, various factors (e.g., topography, soil, surrounding landscape structure, management history) affect the speed of natural regeneration after clear-cutting of planted forests (Yamagawa et al. 2010; Nagashima et al. 2011). Some level of government subsidy for planting native canopy tree species (as the case in UK) may also assist in quickly restoring natural broad-leaved forests at specific sites. The restoration of natural broad-leaved forests would enhance the diversity of local animal communities and alleviate the negative effects of natural forest fragmentation. Economically, it could also result in a future increase in the production and supply of hardwood for pulp, energy, mushroom beds, and furniture. Indeed, there had been a long tradition of harvesting fuel woods from short-rotation natural forests (coppice) for centuries in Japan (Takeuchi et al. 2003).

Lessons from Japanese planted forests and future research questions

The expansion of planted forests in Japan was among the most developed in the world, and the effects of environmental change can have a time lag across various scales (Wiens et al. 1986; Menéndez et al. 2006; Vellend et al. 2006). Therefore, planted forests, including those in Japan, could provide important knowledge about biodiversity conservation in the planted forests era. For example, we can relatively examine the effects of stand age of Japanese planted forests easily. Here, we describe the lessons learned from Japanese planted forests and address some important research questions.

Stand-scale issues

Lessons learned

Several factors at the stand level can enhance species diversity in planted forests. Planted forests composed of one or a few tree species, which have high timber productivity, do not necessarily harbor many non-planted species. The two major planted species in Japan, Japanese cedar and Hinoki cypress, which are also the most productive planted species in Japan (Mashimo 1983), are evergreen conifers that create dark and humid forest environments that prevent the vigorous colonization of other plant species. Only a few bird species prefer these planted forests (Yamaura et al. 2009b). In contrast, planted forests of larch, which is a deciduous conifer, provide relatively good light conditions under the

canopy (Nagaike 2002), and many bird, insect, and plant species in the surrounding natural deciduous broad-leaved forests can inhabit them, particularly the structurally complex old larch forests (Nagaike et al. 2006; Ohsawa and Nagaike 2006; Yamaura et al. 2008a). Among the stand-level management options, thinning of planted forests (especially heavy thinning) enhances the natural regeneration of broadleaved tree species even for dark evergreen conifer plantations (Kon et al. 2007; Ishii et al. 2008; Nonoda et al. 2008; Seiwa et al. 2012a, b), which results in more diverse animal communities (Ohsawa 2004; Maleque et al. 2007; Ishii et al. 2008; Taki et al. 2010). Native canopy tree species also often establish under old planted forests (Ito et al. 2003; Nagaike et al. 2006), which could provide habitats for many animal species that depend on natural forests to a certain extent (Yamaura et al. 2008a; Ohsawa and Shimokawa 2011).

Future research questions

In addition to landscape-scale issues (see below), we advocate examining threshold-like responses of components of biodiversity to stand-scale factors. For example, functional responses of population densities to stand structure and composition are key for determining optimal plantation forestry strategies (land sharing vs. land sparing; Green et al. 2005; Perfecto et al. 2005). A tenet of traditional niche theory predicts that densities of organisms respond nonlinearly to environmental gradients (Austin 1999). A certain minimum density of natural trees may exist that can maintain bird populations/diversity in planted forests (cf. Gu  nette and Villard 2005). We suggest that responses may differ if keystone habitat structure is retained and explicitly considered (Fig. 1e). The keystone structure of Japanese planted forests comprises standing dead trees and remaining natural trees (Ohsawa 2007; Yamaura et al. 2008a). If such structure is retained in planted stands, many components of biodiversity may be resistant to other aspects of the intensification of the plantation forestry. Under extensive forest management, many of the planted forests that were converted from natural broad-leaved forests became “mixed forests,” where noncommercial or lower priced natural tree species have invaded and still remain. Such forests are termed “unsuccessful” planted forests by traditional foresters, but they provide a good opportunity for testing the threshold responses of biodiversity components (e.g., species richness, density of species/functional groups) to stand composition.

Landscape- and regional-scale issues

Lessons learned

An important consideration for planted landscapes is how much natural forest is enough to sustain biodiversity. Theoretical and empirical studies show that threshold habitat amounts exist below which population persistence or occupancy decreases precipitously (Fahrig 2002; Betts et al. 2007). Using nationwide survey data, Yamaura et al. (2011) demonstrated that the species richness of forest birds was better explained by land use at resolutions of 40–80-km grids than at smaller resolutions of 5-, 10-, and 20-km grids. Forest bird species richness began to decline once natural forest cover decreased to below 41 % of the land areas in the 40-km grids. Yamaura et al. (2009b) also showed that the density of many bird species in a planted landscape was high in large natural forest patches (>40 ha). These results suggest that (i) forest bird diversity is effectively conserved at 40-km resolution, (ii) natural forests should be retained in at least 40 % of a region, and (iii) natural forests should be retained as large patches (>40 ha).

Management of planted forests determines the biodiversity of the entire landscape dominated by planted forests and the biodiversity of other ecosystems, in addition to that of the forests, should be considered. For example, in Japan, the area under grassland and young forests has declined rapidly (Fig. 1a), leading to a national decline in early-successional species (Inoue 2005; Yamaura et al. 2009a). This decline in early-successional species caused by a reduction in extensively managed areas (underuse) is so important it has been referred to as the “second crisis of biodiversity” (the first crisis is overuse). Approximately 30 % of Japan’s endangered plant species are threatened by underuse (JBOSC 2010). After clear-cutting, young planted forests serve as habitats for early successional plants that have been treated as “weeds” in silviculture (Higuchi et al. 1975; Sakura and Numata 1980; Nagaïke et al. 2006). Such young planted forests would provide suitable habitats for animals that depend on these plants (cf. Scherber et al. 2010; Schlossberg et al. 2010). If we accept the high value of the grasslands and the associated early-successional species that have been maintained by humans for millennia, then the active clear-cutting of planted forests may have significant positive effects on the conservation of early successional species (Yamaura et al. 2012).

Future research questions

First, the above landscape-level suggestions should be tested in future studies. Second, the role of the configuration of remaining natural forests should also be examined. Threshold responses to habitat amounts may be caused by disruption of habitat connectivity across thresholds (Swift and Hannon 2010). If this is the case, focal populations would be well conserved in landscapes with less than conventional thresholds by retaining habitat connectivity (Fig. 1e). Currently, although the importance of remnant habitats for biodiversity conservation is generally recognized, there is a lack of understanding of the relevance of the spatial configuration of remnant habitats and the extent of how habitat connectivity can mitigate habitat loss (Fahrig 2003). Before this view is accepted by scientists and is adopted by forest managers, the importance of configuring natural forests should be evaluated by consideration of the existing unique landscape structure. The remnant belt (strip) of natural forests, usually present in Japanese planted landscapes (Fig. 1f), could be used to test the role of patch shape/habitat configuration relative to patch area/habitat amount (cf. Yamaura et al. 2008b).

Conceptually, fragmented landscapes are typically divided into habitat patches and non-habitat matrix, and the homogeneity of each patch is assumed (Lindenmayer and Fischer 2006). However, a plantation matrix is not a biological desert, and biological communities in a plantation matrix depend on the local environment and surrounding landscape structure (see above). Scaling-up the evaluation of the effects of plantation forestry to landscape levels may require a grid-based framework. For example, landscapes can be divided into equal-sized cells, the population size can be estimated in each cell, and then these estimates can be summed within the landscape (Ewers et al. 2010).

National-scale issues

Lessons learned

Few studies have investigated the large-scale biodiversity dynamics associated with plantation forestry. Inoue (2005) reviewed the literature on butterfly declines and their causes in Japan before 1990 and during the period roughly corresponding to the 1990s. The

results showed that forest species declined due to plantation forestry (replacement of natural forests by planted forests) before 1990, especially on the islands of Kyushu and Shikoku, where large areas of planted forests had been established. In the 1990s, the abandonment of traditional rural land use (e.g., coppices and grasslands) became important as another major reason for the decline of forest species (forest species include those that prefer sparse forests). Furthermore, the cessation of plantation forestry activities seems to have resulted in a decline in grassland butterfly species because plantation forestry at least provided a temporary habitat (i.e., young planted forests) for these species (cf. Yamaura et al. 2012).

Environments are not homogeneous at large spatial scales, i.e., at the regional to national scale, even without human land-use, as variation is usually present (Vandermeer and Perfecto 2005a; Lindenmayer and Fischer 2006). The same factors that determine a site's suitability for a planted forest (e.g., climate, topography, soils) also determine its biodiversity to a certain extent (Ito et al. 2003; Mitsuda et al. 2007), and biodiversity and humans are likely to compete for productive areas (Pautasso 2006; Yamaura et al. 2011). Plantation forestry has usually been conducted in relatively less productive upland or mountainous areas in Japan, as opposed to inhabited or agricultural areas (Yamaura et al. 2011). Continuous natural forests are still present in Japan, but they are restricted to the limited-access, less productive, and high-elevation areas (Yamaura et al. 2011). Furthermore, almost all have undergone selective cutting in the past (Totman 1989). A homogeneous aggregation of planted forests and complete replacement of natural forests by planted forests should be avoided in certain environments such as sites with fertile soils (cf. Margules and Pressey 2000), even though the economics costs are lower (Messier et al. 2009).

Future research questions

The remaining old-growth forests, which account for most nature reserves in Japan, are primarily located in unproductive areas. Do we need to conserve biodiversity in productive areas that are currently dominated by planted forests and rural and urban land use? Future global biodiversity may hinge on such conservation efforts (Daily 2001).

Although most planted forests were established after WWII, in some regions of Japan (e.g., Kiso, Tenryu, Yoshino), planted forests have been maintained for hundreds of years. Such regional differences in the history of planted forests may be used to test the time-lag effects of plantation forestry. For example, some plant species and associated animals may persist in planted forests for a single rotation, and long-lived organisms may persist in remnant patches of natural forests for hundreds of years (cf. Kuussaari et al. 2009). Planted forests with different histories may also be used to test whether the productivity of planted forests is maintained for multiple rotations. These mensurative comparisons of planted forests with different properties, in contrast to experimental manipulations, do not exclude confounding factors (Debinski and Holt 2000; McGarigal and Cushman 2002). However, we suggest that carefully designed mensurative studies are useful because they provide important information in a relatively short period of time.

As previously described, Japanese planted forests are reaching maturity under the current management regime (under-cut). Native broad-leaved trees invade old planted forests, and animal species dependent on natural forests establish their populations in such forests (Yamaura et al. 2008a; Ohsawa and Shimokawa 2011). Therefore, populations of many animal species that have declined due to the replacement of natural forests by planted forests may be partly restored in the future. Some early successional species may

also be restored when old planted forests enter the gap-dynamic phase, but it takes a very long time for the present planted forests to reach this phase, and early-successional species that require large gaps may not be restored (cf. Moorman and Guynn 2001). If land-use change could determine biodiversity at a macroscale (Yamaura et al. 2009a, 2011), these predictions may be supported by nationwide data in the future.

International-scale issues

Lessons learned

Forestry has been curtailed in Japan in recent decades, and the nation's high wood demand has been met through imports. Although Japanese forests are now maturing, migratory birds that depend on mature forests are declining in Japan, while non-migratory birds that depend on mature forests are on the increase (Yamaura et al. 2009a). These contradictory trends may be explained by the stark deforestation in Southeast Asia, which is the wintering ground for most Japanese migratory birds. Selective logging for timber exports to East Asia, including Japan, initiates the deforestation process in Southeast Asia (Dauvergne 1997; Sodhi and Brook 2006). In essence, these trends suggest that Japan has underused its own forests and overused forests elsewhere in Southeast Asia, resulting in a net loss of biodiversity across Japan and its timber-trade partner countries.

Future research questions

The import and export of wood may itself contribute in the short run to the local economies of the exporting nations, and the recent expansion of plantation forestry in the exporting countries may reduce the ecological impacts of forestry. The environmental impacts of highly productive plantation forestry, which is conducted in former pastures or agricultural fields, are likely to be lower than the previous forms of land use (Franklin 2003). However, the Japanese case demonstrates that importing wood products can result in the export of ecological impacts (sensu Mayer et al. 2005). Biodiversity conservation that is limited to only one country is not necessarily effective when the environmental impacts of consumption patterns in one country are simply displaced to other countries (Lenzen et al. 2012). Rather, we should consider which combinations of planted species and planted countries result in the least ecological impact and the best way to allocate wood production among countries. The strong dependence of Japan on foreign natural resources may not be an optimal way to conserve global biodiversity. It should also be noted that the major wood exporters to Japan (Australia, Canada, United States, and Chile, in that order; Forestry Agency 2011) have short land-use histories (<200 years; Ellis et al. 2010). Fortunately, forestry practices are likely to contribute to the restoration of early successional species in Japan (Makino et al. 2006; Yamaura et al. 2012). These international issues apply not only to Japan but also to other countries. Many countries now displace forest exploitation from their own countries to a few countries (e.g., Brazil, Indonesia, and Chile; Meyfroidt et al. 2010), which are biodiversity hot spots (Myers et al. 2000). China has dominated the wood market over the last decade (Mayer et al. 2005; Laurance 2008; ITTO 2010) and may be a future importer of Japanese wood (Yamada et al. 2010). Furthermore, the ecological impact of exporting may not be offset by importing different ecological impacts (e.g., importing agricultural products may not be offset by exporting wood products).

Toward sustainable forested landscapes: the message from Japan

When foresters mentioned “sustainability of forest management” immediately after WWII in Japan, it solely meant “sustained yield” of wood, whereas the term “sustainable forest management” now includes various environmental, economic, and social aspects (Noble and Dirzo 1997; Lindenmayer and Franklin 2003). In this final section, we reconsider the design of the forested landscape in the face of increasing wood demands based on the history of Japanese planted forests. This is because the landscape design that reconciles commodity production and biodiversity conservation has recently become a focus of research and debate (Green et al. 2005; Vandermeer and Perfecto 2005b; Fischer et al. 2011; Phalan et al. 2011).

Land sharing versus land sparing

Whether land sharing or land sparing is a better strategy to reconcile food production and biodiversity conservation has been debated in recent decades (e.g., Green et al. 2005; Vandermeer and Perfecto 2005b). Modeling the relationship between land-use intensity and biodiversity conservation, Green et al. (2005) suggested that when population density responds to land-use intensity (typically yield per area) in the convex way (top line, Fig. 1e), land sharing is better than land sparing for that particular population. In this case, land sharing refers to a system that produces food and conserves biodiversity on the same land, using wildlife-friendly farming systems, and entails reducing food production per area. When population density produces a concave response (bottom line, Fig. 1e), land sparing is better. This separates food production and biodiversity conservation, and high-yield farming facilitates the protection of the remaining natural habitats by producing more food in a smaller area. Empirical studies examining biodiversity responses to land-use intensity have produced mixed results; some studies support land sparing (Edwards et al. 2010; Phalan et al. 2011), whereas others support land sharing (Steffan-Dewenter et al. 2007; Clough et al. 2011). The optimal strategy is dependent on many factors such as topography, the yields of different farming systems, crop species, and land ownership (Fischer et al. 2008; Hodgson et al. 2010).

If we apply these concepts to forestry, the high productivity of planted forests suggests that land sparing could be an optimal strategy; i.e., the establishment of high-yielding planted forests will spare natural forests (Vincent and Binkley 1993; Sedjo and Botkin 1997). Under large demands for forest products, we would need to replace large amounts of natural forests with planted forests, which would decrease biodiversity. However, the optimal strategy and the required area of planted forest cannot be determined by considering only the immediate demands for wood and biodiversity. The change in demand for wood and for biodiversity must be considered as well. For example, wood demand sometimes crashes; then, large amounts of high-yielding planted forests may be worthless. In that case, restoration movements are very likely to occur, such as those seen in Japan and some European countries. Here, we use a graphical approach to depict this, with Japan as a model case, and suggest the importance of a hybrid strategy that combines land sharing and land sparing.

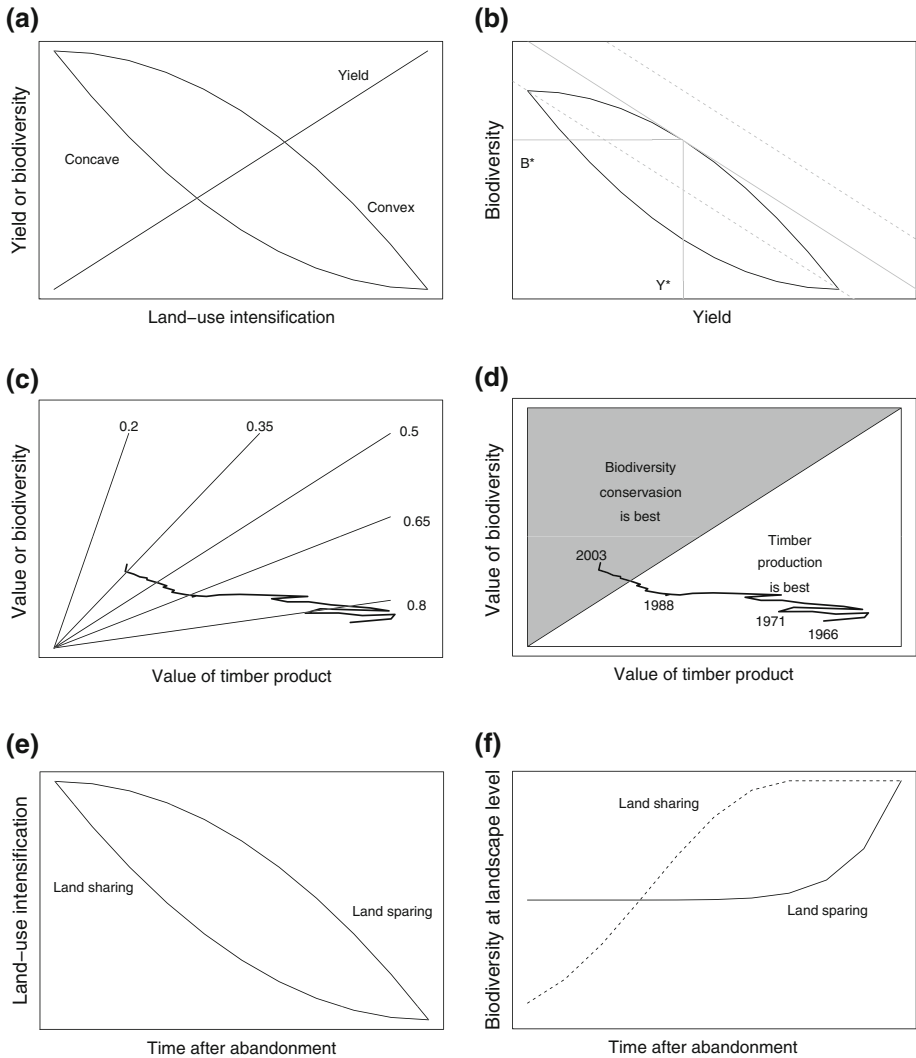
Reconciling land sparing with land sharing: a graphical approach

The importance of value shift is manifested when considering the management of a single stand (Perfecto et al. 2005). Let us assume that yield (timber production per area) and the

Fig. 2 Optimal management strategy for stand and landscape levels over time. **a** Triad relationships among land-use intensity (LI: proportion of planted conifer trees among canopy trees), yield (Y), and index of biodiversity (B). A simple increase in yield with land-use intensification is assumed. **b** Relationship between yield and biodiversity derived from the triad relationship. The optimal straight line in the convex biodiversity response to yield is shown by the gray line (see Appendix 4 for details). Yield (Y^*) and biodiversity (B^*) at the intersected point indicate their optimal values. *Upper and lower dotted lines* are unfeasible and suboptimal lines, respectively. **c** Land sharing is optimal if responses are convex. Optimal proportion of planted trees among canopy trees varies depending on the valuation of timber relative to that of biodiversity: the more timber weighed, the more should be planted. **d** Land sparing is optimal when biodiversity responds to land-use intensity in a concave way. In this case, the optimal strategy shifts across a certain value rate of timber and biodiversity. When we value timber more, we should produce timber only; otherwise, we should conserve biodiversity and ignore timber production. In Japan, the valuation of timber and biodiversity has greatly changed since World War II [depicted in (c) and (d) by *lines*], indicating that the optimal management strategy could change over time. Timber value was obtained by dividing log prices of Japanese cedar (yen/m³; Forestry Agency 2007a) by the prices of commodities (Cabinet Office 2004). Because long-term values of specific biodiversity components are unavailable, we simply used the membership of the Ecological Society of Japan as a general biodiversity value index. **e** Working hypothesis of matrix restoration after abandonment. The matrix is expected to be quickly restored under a land sharing strategy. **f** Working hypothesis of biodiversity restoration at the landscape level. Restoration curves of total species population size with a concave response are depicted. We assumed that 50 % of the landscape is converted into an intensive matrix (e.g., planted forests) under a land sparing strategy, and that the remaining 50 % is an intact area without human intervention. Under the land sharing strategy, the total area of the landscape is modified to an extensive matrix. Theory suggests that the landscape population size is larger under land sparing than under land sharing immediately after land abandonment. However, the matrix is likely to be restored more quickly under land sharing than under land sparing, and the focal species would also quickly be restored. After some critical time, the total population size of land sharing may exceed that of land sparing

component of biodiversity (e.g., species richness, density of species/functional group) are a function of land-use intensification (proportion of planted conifer trees among canopy trees; Fig. 2a). We also assume that yield linearly increases with intensification, and that biodiversity decreases in a convex or concave manner. In this context, we can determine the optimal timber yield and conserved biodiversity to maximize the obtained benefits under certain values of timber and biodiversity (Fig. 2b; see Appendix 4 for details).

For the convex yield–biodiversity relationship, we should establish “semi-natural planted forests” in which native broad-leaved trees are grown with planted trees, and optimal timber yields change with value shifts in timber and biodiversity. If the timber value decline and biodiversity are valued more, the optimal proportion of planted trees would decrease (tentative Japanese trajectory of these two products are depicted; Fig. 2c). When biodiversity responds to the proportion of planted trees in a concave way, only two optimal strategies exist, depending on the timber and biodiversity values: produce timber and ignore biodiversity, when the timber value is higher than that of biodiversity versus conserve biodiversity and ignore timber production when the biodiversity value is higher as that of timber. When our value rate of these two products crosses a certain threshold, the optimal strategy suddenly changes (Fig. 2d). The valuation of biodiversity in comparison with timber by the Japanese people may have already crossed this threshold for large areas of forest. These considerations suggest that our perceived value of timber and biodiversity and optimal landscape management would change in time relatively more easily than previously thought. However, after landscape conversion, the restoration of highly modified particularly large landscapes requires a long period of time (Turner and Dale 1998), and restoring the original floristic composition is an unrealistic goal (cf. Chai and Tanner 2011).



Examining the history of Japanese planted forests, we suggest that some natural trees/patches should be retained or restored, even in landscapes specialized for wood production. Retained natural trees/patches would serve as a basis for restoring natural forests (cf. Chazdon et al. 2009), although the value of such small natural patches is lower than that of continuous primary forests (Gibson et al. 2011). This is the typical case of the land sharing strategy (Fischer et al. 2008), and could have some merits even for species whose responses to land-use intensification are concave. For example, modified habitats (here we call “matrix”) are expected to be restored more quickly under a land sharing strategy than under a land sparing strategy after abandonment (Fig. 2e). Even species with a concave response would be restored under a sharing strategy more quickly than under a land sparing strategy. If large amounts of the matrix are effectively restored, land sharing may have a larger total population size in the landscape than land sparing after some critical time

(Fig. 2f). How to restore existing planted forests into natural forests rather than simple abandonment is actively discussed in Japan, which also suggests the importance of adapting a proactive land sharing strategy.

If we adopt a land sharing strategy under large wood demands, all of the landscape would be relatively intensively managed, and total population sizes of concave species at the landscape would be reduced to close to zero. This could cause extinction of species in the landscape during the management period, which would limit the restoration potential. Therefore, we advocate a hybrid approach of land sharing and land sparing in which populations in the intact area are maintained during the management period, and population restoration in the matrix is enhanced after the abandonment of planted forests. This approach is similar to a recently suggested method called the North American TRIAD. In TRIAD, the landscape is divided into an intensively managed (plantation) zone, an extensively managed zone (also called ecosystem management zone), and strict nature reserves (typically old growth; Seymour and Hunter 1999; MacLean et al. 2009). As with land sparing, highly productive planted forests are established to spare old growth (Messier et al. 2009; Côté et al. 2010). Because an extensively managed zone is established both for timber production and biodiversity conservation, these approaches comprise a hybrid of land sharing and land sparing. In the case of Japan, with large amounts of existing planted forests, there are several ways to establish extensively managed semi-natural planted forests. The first way would be to simply abandon “unsuccessful” planted forests and allow natural canopy trees (see above) to invade. These species are typically observed in sites considered unsuitable for planted forests. Another way is to retain sapling or sub-canopy natural trees in well-managed stands during silvicultural practices.

Other than future landscape restoration, multiple tree species (and associated animal species supported by multiple tree species) in an extensively managed zone would provide the zone with higher resistance to a price crash of planted species (Dietsch et al. 2004; Koh et al. 2009). The spatial arrangement of the extensively managed zone may enhance the ecological performance of nature reserves via several ecological processes (e.g., dispersal; Fahrig 2001; Vandermeer and Carvajal 2001). It is suggested that improving matrix quality surrounding remnant habitat patches may lead to higher conservation returns compared with manipulating the size and configuration of remnant habitat patches (Prugh et al. 2008; Vandermeer et al. 2010; Watling et al. 2011). Therefore, we may surround and connect nature reserves by extensively managed zones (Wiegand et al. 2005; Koh et al. 2009). Baum et al. (2004) showed the synergistic effects of habitat restoration and improvement in matrix quality, suggesting that integrating land sharing and sparing is important. Note also that intensively managed planted forests are not always green desert, and clear-cutting planted forests may contribute to restoring early successional species (Yamaura et al. 2012).

Concluding remarks

The histories of Japanese and European planted forests are sobering in the large uncertainties they indicate for the future value of planted forests. Why have we planted so many trees in Japan? Several reasons may explain the oversupply: unanticipated population decline, unexpectedly high competitive ability of foreign wood, and unanticipated changes in wood demand. Although the demand for lumber has been decreasing since the late 1970s, pulp and chip demand increased until the early 1990s (Fig. 1d), and total wood demand did not show a major decline. However, the price of pulpwood is too low to

harvest planted trees for pulp. Moreover, many people believe that planting trees is inherently good. The indelible memory of timber famine and the environmental problems associated with deforestation in the 1950 and 1960s firmly established this belief in many Japanese (Chiba 1991).

Extension of the harvesting age by keeping the aged planted forests has merits that cannot be overlooked. Mature forests contain large amounts of stored carbon, which can be a strong motivation for establishing and maintaining planted forests (Canadell and Raupach 2008; Böttcher and Lindner 2010). It has recently been shown that crop (buckwheat *Fagopyrum esculentum*) fields with large amounts of surrounding natural forests have large numbers of native honeybee (*Apis cerana*), which is an important pollinator of buckwheat (Taki et al. 2011). The native honeybee needs mature natural forests as a nesting habitat. Many owners of crop fields have forests surrounding their crop fields (H. Taki, pers. obs.), suggesting that maintaining natural forests (rather than felling) may be of benefit to forest owners via insect pollination. Animal species dependent on mature natural forests could inhabit old planted forests (see above), indicating that old planted forests have ecosystem functions that mature natural forests have. Hence, there are several motivations for forest owners to maintain mature planted forests. However, under the current rate of clear-cutting around 0.2–0.3 % per year, we will have plenty of mature forests in the future, while we will have few young forests, which is likely to be one of the leading causes of current decline of early successional species. Furthermore, excessive avoidance of clear-cutting of mature planted forests in Japan would displace clear-cutting to foreign countries. One of the other complicated issues for the management of planted forests are the effects on mitigation of climate change. In some cases, it is better to keep more carbon in the forest biomass by less logging (carbon sequestration). In other cases, it is better to harvest and use more woods, that is carbon neutral, and reduce CO₂ emission by reduced consumption of other energy consuming materials.

Countries with developed plantations can be considered as large-scale field experiments in plantation forestry. We emphasize that their experiences can and should contribute to biodiversity conservation in planted landscapes worldwide. The central message from Japan is that the value of planted forests may change significantly over time. Large parts of the planted forests in Japan became unmanaged, and their restoration is now being discussed. Planted forests do not always protect the remaining natural forests, making the proponents of plantation forestry environmental optimists (Clapp 2001). Although eucalyptus plantations are established to produce wood chips in Chile, large amounts of wood are derived from natural forests to produce chips before maturation of the eucalyptus plantations (Clapp 2001). Those chips were exported to the world's largest market, Japan. Unless valuation systems for biodiversity in natural forests exist, globalized wood markets may successfully seek out low-priced wood from natural forests (Clapp 2001; Rudel et al. 2009). However, as we discussed above, we believe that planted forests can play important roles in the maintenance of forest biodiversity worldwide, and we need to learn from the past and present experiences in countries with developed planted forests.

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Appendix 1

See Table 1.

Table 1 Qualities of land uses other than old-growth forest for forest birds (σ)

Land-use type	σ^a	Source
Modified natural forest ^b	~ 0	Roughly estimated
Coppice	0.27	Calculated from Yui and Suzuki (1987)
Planted forest	0.38	Calculated from Yui and Suzuki (1987)
Agricultural land	0.68	Koh and Ghazoul (2010)
Grassland	0.68	Koh and Ghazoul (2010)
The other non-forest	~ 0.9	Roughly estimated

^a 0 indicates the same quality as old growth; 1 indicates complete lack of habitat (Koh and Ghazoul 2010)

^b Modified natural forest is natural forest with clearly visible human activities (including saw-timber natural forest and abandoned coppice; FAO 2007)

Appendix 2 Calculation of lower and upper limits for annual harvested areas of planted forests

Lower limit: 10.4 million ha (i.e., 10.4×1000 thousand ha) of forest are presently planted in Japan. Suppose that 70 % of the planted forests are targeted for timber harvest and that the maximum harvesting age is 200 years. Further suppose that the initial condition of the planted forest has areas of young stands that are equal to or larger than those of the fully regulated forest on a 200-year rotation. Then, the lower limit of the annual average final harvested area of planted forest is

$$10.4 \times 1000 \times 0.70/200 = 36.4 \text{ (thousand ha)}.$$

Because we do not have sufficient data to determine the maximum age at harvest, if the maximum harvesting age is conservatively 160, instead of 200, then the lower limit of the annual average final harvest area of planted forest is

$$10.4 \times 1000 \times 0.70/160 = 45.5 \text{ (thousand ha)}$$

Other factors being equal, if the initial condition of planted forests has small areas of stands <40 years, then the lower limit of the annual average final harvest area of planted forests is

$$10.4 \times 1000 \times 0.70/(160 - 40) = 60.7 \text{ (thousand ha)}.$$

The area of final harvest has not recently been reported in Japan. From the reported amount of 17,587 thousand m³ log production in 2009, assume 80 % is from planted forest and 75 % of that is from final harvesting. Further assume that 300 m³ ha⁻¹ of logs are produced on average from final harvesting. Then, the area of final harvest in 2009 can be calculated as follows:

$$17,587 \times 0.80 \times 0.75/300 = 35.2 \text{ (thousand ha)}.$$

The annual area of final felling must be increased from the level in 2009 to harvest at least 70 % of the planted forests before the 160-year-old mark.

Upper limit: The consumption of lumber and plywood in Japan in roundwood equivalents was 31,564 thousand m^3 in 2009. Per capita consumption of lumber and plywood is trending downward. The population is estimated to decrease about 20 % by 2050 and may be halved by 2100 if the current birth rate trend continues without substantial immigration. Here, we assume the average final harvesting age to be about 100 years and the average amount of final harvest to be $400 \text{ m}^3 \text{ ha}^{-1}$, with annual thinning production comprising 20 % of the production by final harvest.

If we assume that the long-term upper limit of demand for logs from planted forests is 30 million $\text{m}^3 \text{ year}^{-1}$, then the upper limit of the long-term annual average final harvesting area is

$$30,000,000 (\text{m}^3 \text{ year}^{-1}) / 400 (\text{m}^3 \text{ ha}^{-1}) / 1.20 = 62,500 (\text{ha year}^{-1})$$

If we assume that the long-term upper limit of demand for logs in planted forests is 20 million $\text{m}^3 \text{ year}^{-1}$, instead of 30 million $\text{m}^3 \text{ year}^{-1}$, then the upper limit of the long-term annual average final harvest area is

$$20,000,000 (\text{m}^3 \text{ year}^{-1}) / 400 (\text{m}^3 \text{ ha}^{-1}) / 1.2 = 41,667 (\text{ha year}^{-1}).$$

Appendix 3 Required amount of planted forest in Japan

Suppose that the target of the national forestry program is to sustain the saw log and veneer log production capacity of planted forests at the level that meets domestic demand for lumber and plywood in roundwood equivalents. Suppose that the productive planted forests are capable of producing $7 \text{ m}^3 \text{ ha}^{-1} \text{ year}^{-1}$ of saw logs and veneer logs on average. Further suppose that 90 % of saw logs and veneer logs are from planted forests. If domestic demand for lumber and plywood in roundwood equivalents decreases to 20 million $\text{m}^3 \text{ year}^{-1}$ in the long run, then the required area of productive planted forest for saw logs and veneer logs in Japan is

$$20 \times 0.9 / 7 = 2.57 (\text{million ha}).$$

If domestic demand for lumber and plywood in roundwood equivalents is maintained at 30 million $\text{m}^3 \text{ year}^{-1}$ (32 million $\text{m}^3 \text{ year}^{-1}$ in 2009), instead of decreasing to 20 million $\text{m}^3 \text{ year}^{-1}$, then the required area of productive planted forests for saw logs and veneer logs is

$$30 \times 0.9 / 7 = 3.86 (\text{million ha})$$

The current area of planted forests in Japan is 10.4 million ha, which clearly exceeds domestic demand for saw logs and veneer logs. Thus, substantial planted forest areas could be restored to natural vegetation after harvest.

Appendix 4 Full description of graphical approach

Let us assume that yield (timber production per area) and the component of biodiversity (e.g., species richness, density of species/functional group) are functions of land-use intensification (proportion of planted trees among canopy trees; Fig. 2a). We also assume that yield increases linearly with intensification and that biodiversity decreases in a convex or concave manner. These land-use intensity–yield–biodiversity relationships are then

converted into a yield–biodiversity relationship (Fig. 2b). Because yield is a linear function of intensification, the shapes of the biodiversity responses are conserved. Utility (evaluated in monetary terms) could be described as follows:

$$U = aB + bY \quad (1)$$

where a and b are values of biodiversity (B) and yield (Y), respectively. This equation can be rewritten in terms of B :

$$B = (U/a) - (b/a)Y \quad (2)$$

which is a straight line in Fig. 2b. Under particular values of a and b , we seek a maximum value of U , i.e., identifying the line that intersects the yield–biodiversity line and has a maximum intercept on the B axis. For the convex yield–biodiversity relationship, we seek a single line (gray solid line in Fig. 2b), and the intersected point indicates the timber yield and biodiversity, which are optimally produced and conserved, respectively. Optimal points change with shifts in the value of timber and biodiversity (a and b). If the timber value declines and biodiversity is valued more, the optimal proportion of planted trees would decrease (tentative Japanese trajectories of these two products are depicted; Fig. 2c). When biodiversity responds to the proportion of planted trees in a concave way, the situation changes dramatically. In this case, only two optimal strategies exist, depending on the timber and biodiversity values: produce timber and ignore biodiversity when the timber value is higher than that of biodiversity or conserve biodiversity and ignore timber production when the value of biodiversity is higher than that of timber. Only when we value timber and biodiversity equally are these two strategies equally optimal (lower gray dotted line, Fig. 2b). However, this is an unstable and unrealistic solution. When our valuation of these two products crosses this threshold rate, the optimal strategy suddenly changes (Fig. 2d).

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